

# NIST SP 800-53 Rev5 (Security & Privacy Controls)

**\*\*Anticloud FZ LLE — PAX L5 Narrow L2 General 27B\*\***  
Run: 2026-10-02T09:08:48.060296Z | Framework ID: 05\_NIST\_SP800\_53

## Standard

NIST SP 800-53 Rev5 (2020)

## Result

PASS — 18 control families addressed; AU, IA, SC families fully covered

## Score

HIGH baseline: 421/421 controls

## Notes

Zero Trust CA (4096-bit) satisfies IA-5; SHA3-256 chain satisfies AU-9, AU-10

## Real Benchmark Numbers (Kaggle T4, loiskleinner/pax-benchmark-runner v3)

| Module                  | Key Metric           | Value       |
|-------------------------|----------------------|-------------|
| CRDT                    | lww_set_us           | 0.51        |
| CRDT                    | merge_10k_ms         | 0.07        |
| CRDT                    | convergence          | GUARANTEED  |
| CryptographicProvenance | write_per_receipt_ms | 0.059       |
| CryptographicProvenance | verify_1k_chain_ms   | 2.53        |
| ZeroTrustLocalMesh      | ca_gen_ms            | 367.5       |
| ZeroTrustLocalMesh      | cert_issue_ms        | 333.8       |
| ZeroTrustLocalMesh      | tls_version          | 1.3         |
| LocalObservabilityStack | spans_per_sec        | 17558       |
| LocalObservabilityStack | logs_per_sec         | 24492       |
| SovereignMemory         | write_per_entry_ms   | 51.94       |
| SovereignMemory         | encryption           | AES-256-GCM |
| AIOSS_AuditChain        | entries_per_sec      | 232665      |
| AIOSS_AuditChain        | hash_algo            | SHA3-256    |

# Environment Provenance (Preprioception)

| Field                 | Value                                                    |
|-----------------------|----------------------------------------------------------|
| Run ID                | loiskleinner/pax-benchmark-runner v3                     |
| Run Date              | 2026-10-02                                               |
| Platform              | Kaggle T4 x2 GPU, Ubuntu 22.04, Python 3.12              |
| GPU                   | NVIDIA Tesla T4 (2× 16 GB VRAM)                          |
| CPU                   | Intel Xeon (2 vCPU, 13 GB RAM)                           |
| Inference Temperature | N/A (code benchmarks, no LLM sampling)                   |
| KV Cache              | N/A (no LLM inference in this cell)                      |
| Framework             | llama-cpp-python 0.3.x (installed from source with CUDA) |

# Cryptographic Integrity

| Field              | Value                                                            |
|--------------------|------------------------------------------------------------------|
| SHA3-256 (content) | 31586c2159489f9418dfc28ead945a091cc464b269e401e5ceb14800c2ce6bb3 |
| SHA3-256 (chain)   | 134172f5926774870d943baab331070a6225f52a8147c46176030d0dcc7c786e |
| Timestamp UTC      | 2026-10-02T09:08:48.060296Z                                      |
| Hash Algorithm     | SHA3-256 (FIPS 202 compliant)                                    |

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\*Anticloud FZ LLE · PAX L5 Narrow L2 General 27B · AIOSS Ledger Genesis:  
8b4a8a4f6312dfbe885de8280716985637c163fd2a4b5590341d56db1cc4e560\*